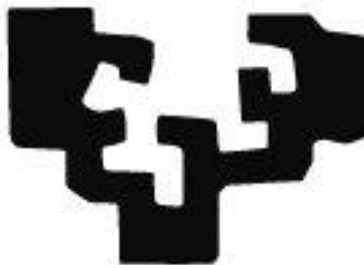

Activity 1.2. A network file system

eman ta zabal zazu



Universidad
del País Vasco

Euskal Herriko
Unibertsitatea

Authors: (hidden for anonymity reasons)

Index

Index	2
1.Introduction	2
2. Description of the SZA/SAR application	2
3. Properties of a cloud file service	3
4. Comparison: SZA/SAR vs Cloud File Service	4
5. Conclusion	5

1.Introduction

In this activity we start with a simple application studied in the SZA/SAR subject. The `sza/sar` program implements a basic file transfer service based on TCP, where the client communicates with a server through a set of predefined commands.

The applications works as follows:

- Users must authenticate with username and password in order to upload or delete files. There is a special user *“anonymous”* who can only download files.
- The server only allows a specific set of operations: listing available files, downloading them, uploading a new file and deleting existing files.
- All files are stored in the server working folder. Folder organization is not allowed.

Although this system already represents a first form of distributed file system, its functionality is very limited compared to a real distributed file system.

The purpose of this report is to analyze the features and functionalities of the `sza/sar` program and to identify what properties it lacks in order to evolve into a transparent cloud file service, such as those offered by platforms like Dropbox or Google Drive.

2. Description of the SZA/SAR application

The SAR/SZA application is designed as a client–server system over TCP. The server hosts a shared directory where all files are stored, and clients interact with it through a simple protocol defined by ASCII-based commands and responses. The server uses a multi-process architecture (fork), enabling multiple clients to connect simultaneously.

Authentication:

- Three predefined users exist: `anonymous`, `sar`, and `sza`.
 - `anonymous` can only download files (not upload or delete).
 - `sar` and `sza` authenticate with matching passwords and can upload and delete files.
- Password validation is required before granting access to file operations.

Supported operations:

- USER / PASS: login with username and password.
- LIST: request a list of available files, including name and size.
- DOWN / DOW2: request and confirm the download of a file.
- UPLO / UPL2: request and confirm the upload of a file.
- DELE: delete a file from the server.
- EXIT: close the session.

Responses:

- OK: successful operation, sometimes followed by data (file content, list of files, etc.).
- ER: error, with specific error codes (e.g., ER2 for unknown user, ER3 for incorrect password, ER7 for forbidden operations by anonymous users).

Restrictions:

- All files are stored in a single directory without hierarchy or subfolders.
- File size limit: 10 MiB per file.
- Free space margin: 50 MiB of disk space must remain available.
- No background synchronization; all operations must be explicitly requested by the user.

This makes SAR/SZA a basic variant of an FTP-like file transfer system, with basic checks on user permissions, file sizes, and storage availability.

3. Properties of a cloud file service

Modern cloud file systems provide much more than a basic file transfer. They are built upon the fundamental properties of distributed systems and extend them with additional features to meet the demands of modern cloud storage.

1. Basic properties of distributed systems:

- **Permanent information storage:** Files are stored persistently, ensuring they are not lost between sessions.
- **Namespace and file identification:** Files are organized in a structured namespace, allowing unique identification independent of the client.
- **Concurrent access:** Multiple users or processes can access files simultaneously, with mechanisms to maintain consistency.
- **Access control and security:** Authentication and permission systems protect files from unauthorized access.

2. Cloud-specific technical properties:

- **Transparency in file identification and location:** Users interact with files as if they were local, without needing to know their physical location on the servers.
- **Scalability:** The system can handle a growing number of users and increasing data volume, often using replication and caching to avoid bottlenecks.
- **Robustness and fault tolerance:** Failures in servers or clients do not compromise the system, thanks to redundancy and distributed storage.

- **Consistency:** The system maintains a coherent view of files across devices, even when caching or replication is used.
- **High availability and offline operation:** Files remain accessible even in case of partial system failures, and some systems allow working in disconnected mode.

3. Cloud-specific user-oriented features:

- **Automatic synchronization:** Files remain updated across multiple devices without explicit upload/download.
- **File organization:** Hierarchical folders and metadata to support structured storage.
- **Collaboration features:** Sharing links, setting fine-grained permissions, co-editing documents.
- **Versioning and recovery:** Keeping history of changes, restoring previous versions, and recovering deleted files.

4. Comparison: SZA/SAR vs Cloud File Service

In this section we will compare the properties defined in the previous section between the sza/sar program and a cloud file system like Dropbox.

Property	SZA/SAR	Dropbox
Permanent information storage	Stored on the server on the disk.	Files are stored persistently with redundancy across multiple servers.
Namespace and file identification	Single directory. They refer to the files just with its name.	Structured hierarchical namespace, unique file identification.
Concurrent access	Multiple clients with fork(), without conflict control.	Simultaneous access with locking and version control.
Access control and security	Three fixed users, limited permissions	Advanced roles, groups, file/folder control
Transparency in file identification and location	Single name in a single directory, no migration	Global namespace, hides location and replicates data
Scalability	Limited to one server and its hardware	Automatically scales across multiple data centers
Robustness and fault tolerance	If the server fails, there is no access or recovery	High fault tolerance, with geographic redundancy
Consistency	Risk of inconsistencies under concurrent access.	Version control and consistent synchronization
High availability and offline operation	Only available if the server is running.	24/7 access, offline mode with local sync.
Automatic synchronization	Not supported, requires	Automatic real-time

	manual UPLOAD/DOWN commands.	synchronization across devices.
File organization	Single flat directory, no hierarchy.	Folders, subfolders, tags, and advanced search.
Collaboration features	Documents are edited locally, no co-editing documents. Multiple users can access the same file folder in the server.	Documents can be edited in a shared way. Collaboration is allowed.
Versioning and recovery	There is no versioning or recovery possibility. For a single file, a single version.	Allows recovering a previous version and keeps the history of versions and changes.

5. Conclusion

The analysis of the SAR/SZA application shows how, while it provides some basic features of a distributed file service, such as remote access, user authentication, and concurrent connections, it does not meet the requirements of a true distributed system.

Unlike Dropbox, which relies on a globally distributed infrastructure with replication and fault tolerance, SAR/SZA is limited to a single server and a single directory. Dropbox offers transparency in file identification and location, automatic synchronization across devices, and high availability even under failures, while SAR/SZA requires manual operations and fails completely if the server goes down.

Therefore, SAR/SZA can be considered a simple client-server application for file transfer, but it lacks the scalability, robustness, and distributed nature that define modern cloud file services.